

Derived Logical Connectives And Examples

Example 1

- Suppose we have two simple statements: P, Q
- Form a compound statement using AND logical connectives
- Answer: $p \wedge q$

p	q	$p \wedge q$
F	F	F
F	T	F
T	F	F
T	T	T

- Suppose we have two simple statements: P, Q
- Form a compound statement using AND logical connectives and negate it
- Answer: $\sim (p \wedge q)$

p	q	$p \wedge q$	$\sim(p \wedge q)$
F	F	F	T
F	T	F	T
T	F	F	T
T	T	T	F

Example 2

- Suppose we have two simple statements: P, Q
- Form a compound statement using OR logical connectives
- Answer: $p \vee q$

p	q	$p \vee q$
F	F	F
F	T	T
T	F	T
T	T	T

- Suppose we have two simple statements: P, Q
- Form a compound statement using OR logical connectives and negate it
- Answer: $\sim (p \vee q)$

p	q	$p \vee q$	$\sim(p \vee q)$
F	F	F	T
F	T	T	F
T	F	T	F
T	T	T	F

NAND (Not And)

- It means negation after AND operation of two statements. Assume p and q be two propositions. The NAND operation of p and q to be a proposition which is false when both p and q are true, otherwise true. It is denoted by $p \uparrow q$.

p	q	$p \uparrow q$
T	T	F
T	F	T
F	T	T
F	F	T

NOR (Not OR)

- It means negation after OR operation of two statements. Assume p and q be two propositions. The NOR operation of p and q to be a proposition which is true when both p and q are false, otherwise false. It is denoted by $p \downarrow q$. It is also known as the Joint Denial.

p	q	$p \downarrow q$
T	T	F
T	F	F
F	T	F
F	F	T

XOR (Exclusive OR)

- Assume p and q be two propositions. The XOR operation of p and q is true if p is true or q is true but not both and vice-versa. It is denoted by $p \oplus q$.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

XNOR (Exclusive Not OR)

- Assume p and q be two propositions. The XNOR operation of p and q is true if both of them are true or false. It is denoted by $P \oplus Q$.

P	Q	$\overline{P \oplus Q}$
T	T	T
T	F	F
F	T	F
F	F	T

Example 3: Prove that $X \oplus Y \cong (X \wedge \sim Y) \vee (\sim X \wedge Y)$

X	Y	$X \oplus Y$	$\sim Y$	$\sim X$	$X \wedge \sim Y$	$\sim X \wedge Y$	$(X \wedge \sim Y) \vee (\sim X \wedge Y)$
T	T	F	F	F	F	F	F
T	F	T	T	F	T	F	T
F	T	T	F	T	F	T	T
F	F	F	T	T	F	F	F

As the truth table for both the proposition is the same $X \oplus Y \cong (X \wedge \sim Y) \vee (\sim X \wedge Y)$. Hence Proved.

Example 4: Prove that $(P \oplus Q) \vee (P \downarrow Q)$ is equivalent to $P \uparrow Q$

P	Q	$P \oplus Q$	$P \downarrow Q$	$P \uparrow Q$	$(P \oplus Q) \vee (P \downarrow Q)$
T	T	F	F	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	T	T	T

As the truth table for both the proposition are the same, $(P \oplus Q) \vee (P \downarrow Q)$ is equivalent to $P \uparrow Q$ [Proved]

Variations in Conditional Statement

- Contrapositive: The proposition $\sim q \rightarrow \sim p$ is called contrapositive of $p \rightarrow q$.
- Converse: The proposition $q \rightarrow p$ is called the converse of $p \rightarrow q$.
- Inverse: The proposition $\sim p \rightarrow \sim q$ is called the inverse of $p \rightarrow q$.

Example 5: Show that $P \rightarrow Q$ and its contrapositive $\sim Q \rightarrow \sim P$ are logically equivalent

P	Q	$\sim P$	$\sim Q$	$P \rightarrow Q$	$\sim Q \rightarrow \sim P$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

As the truth table for both the proposition are the same, $P \rightarrow Q$ and its contrapositive $\sim Q \rightarrow \sim P$ are logically equivalent [Showed]

Example 6: Show that proposition $Q \rightarrow P$, and $\sim P \rightarrow \sim Q$ are not equivalent to $P \rightarrow Q$.

P	Q	$\sim P$	$\sim Q$	$P \rightarrow Q$	$\sim P \rightarrow \sim Q$	$Q \rightarrow P$	$(Q \rightarrow P) \wedge \sim P \rightarrow \sim Q$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	T
F	T	T	F	T	F	F	F
F	F	T	T	T	T	T	T

As the truth table for both the proposition are not the same, $q \rightarrow p$, and $\sim p \rightarrow \sim q$ are not equivalent to $p \rightarrow q$ [Showed]

Example 6: Prove that $P \longleftrightarrow Q$ is equivalent to $(P \rightarrow Q) \wedge (Q \rightarrow P)$

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$	$P \longleftrightarrow Q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

As the truth table for both the proposition are the same, $P \longleftrightarrow Q$ is equivalent to $(P \rightarrow Q) \wedge (Q \rightarrow P)$ [Shown]

Example 6: Find the truth table of $\sim P \vee Q$.

P	Q	$\sim P$	$\sim P \vee Q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T